Class 10 Science - Full Detailed Notes with Definitions

1. Chemical Reactions and Equations

Chemical Reaction: A process in which substances change into new substances with different properties.

Types:
• Combination: Two or more substances combine to form one product.
• Decomposition: One substance breaks into simpler substances.
• Displacement: One element replaces another.
• Double Displacement: Exchange of ions between compounds.

Balancing Equation: Making equal number of atoms on both sides of equation.
Oxidation: Gain of oxygen or loss of hydrogen.
Reduction: Loss of oxygen or gain of hydrogen.

2. Acids, Bases and Salts

Acid: Substance that releases H^+ ions in water.
Base: Substance that releases OH^- ions in water.
pH Scale: Measures acidity or basicity from 0 to 14.
Neutralization: Reaction between acid and base to form salt and water.
Indicators: Substances that change color in acids or bases.

3. Metals and Non-Metals

Metals: Elements that are malleable, ductile and good conductors of heat and electricity.
Non-Metals: Elements that are brittle and poor conductors.
Reactivity Series: Arrangement of metals based on reactivity.
Ionic Bond: Chemical bond formed by transfer of electrons.

4. Carbon and its Compounds

Carbon: Tetravalent element forming strong covalent bonds.
Covalent Bond: Sharing of electrons between atoms.
Hydrocarbons: Compounds made of carbon and hydrogen.
Soaps: Cleaning agents made from fats and oils.

5. Life Processes

Life Processes: Basic processes necessary for survival of living organisms.

Nutrition: Process of obtaining and using food.
Respiration: Process of releasing energy from food.
Transportation: Movement of substances inside organism.
Excretion: Removal of waste products from body.

In Humans:
• Nutrition: Digestion in stomach and intestines.
• Respiration: Oxygen used to release energy.
• Transport: Blood carries oxygen and nutrients.
• Excretion: Kidneys remove waste.

6. Control and Coordination

Control: Regulation of body activities.
Coordination: Working together of different body parts.
Nervous System: Brain, spinal cord and nerves.
Hormones: Chemical messengers released by glands.
Reflex Action: Automatic response to stimulus.

7. Reproduction

Reproduction: Process of producing new individuals.
Asexual: One parent involved.
Sexual: Two parents involved.
Fertilization: Fusion of male and female gametes.
Human Reproductive System: Organs involved in reproduction.

8. Heredity and Evolution

Heredity: Transmission of traits from parents to offspring.
Evolution: Gradual change in organisms over time.
Mendel’s Laws: Laws explaining inheritance of traits.
Traits: Characteristics passed through genes.

9. Light

Light: Form of energy that enables vision.
Reflection: Bouncing back of light from surface.
Refraction: Bending of light when passing through medium.
Mirror Formula: Relation between object distance, image distance and focal length.

10. Human Eye

Human Eye: Optical instrument for vision.
Defects: Myopia, Hypermetropia.
Dispersion: Splitting of white light into colors.

Scattering: Spreading of light in different directions.

11. Electricity

Electricity: Flow of electric charge.
Ohm’s Law: V = IR.
Resistance: Opposition to flow of current.
Series Circuit: Components connected in a single path.
Parallel Circuit: Multiple paths for current.

12. Magnetic Effects of Current

Magnetic Field: Region around magnet where force is experienced.
Electromagnet: Temporary magnet using electric current.
Motor: Converts electrical energy into mechanical energy.
Generator: Converts mechanical energy into electrical energy.